Monty Hall Shock Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 30-40 min

Category	Mathematics	**Time**	30-40 min
Difficulty	Medium	**ID**	monty-hall-shock

Big idea: Formal probability can beat intuition.

Student-Facing Prompt

Inspect Monty Hall Shock Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Monty Hall Shock Stimulus A	Student-facing stimulus 1: Three paper doors, one prize card, and repeated rounds with switch/stay tracking. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Monty Hall Shock Stimulus B	Student-facing stimulus 2: A class results table showing wins after switching versus staying. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Monty Hall Shock Reveal / Twist	Student-facing stimulus 3: A probability tree showing that the initial choice has probability 1/3 and the two unchosen doors together have probability 2/3. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Formal probability can beat intuition.

Actual Lesson Questions

#	Question for students
1	After inspecting Monty Hall Shock Stimulus A, what is your first knowledge claim?
2	Which exact detail from Monty Hall Shock Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Why does mathematical reasoning often feel wrong at first?

Ready-to-Use Examples

• Three paper doors, one prize card, and repeated rounds with switch/stay tracking.

• A class results table showing wins after switching versus staying.

• A probability tree showing that the initial choice has probability 1/3 and the two unchosen doors together have probability 2/3.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

1. Play the Monty Hall game physically. A student chooses a door.
2. The host opens an empty door.
3. The student chooses whether to stay or switch.
4. Repeat many times and compare win rates for stay and switch.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Why does mathematical reasoning often feel wrong at first?
- Should we trust intuition or formal proof?
- What makes a mathematical explanation convincing?

Knowledge Questions

- Why does mathematical reasoning often feel wrong at first?
- Should we trust intuition or formal proof?
- What makes a mathematical explanation convincing?

Transfer Example

For example, run this activity with three paper doors, one prize card, and repeated rounds with switch/stay tracking. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...